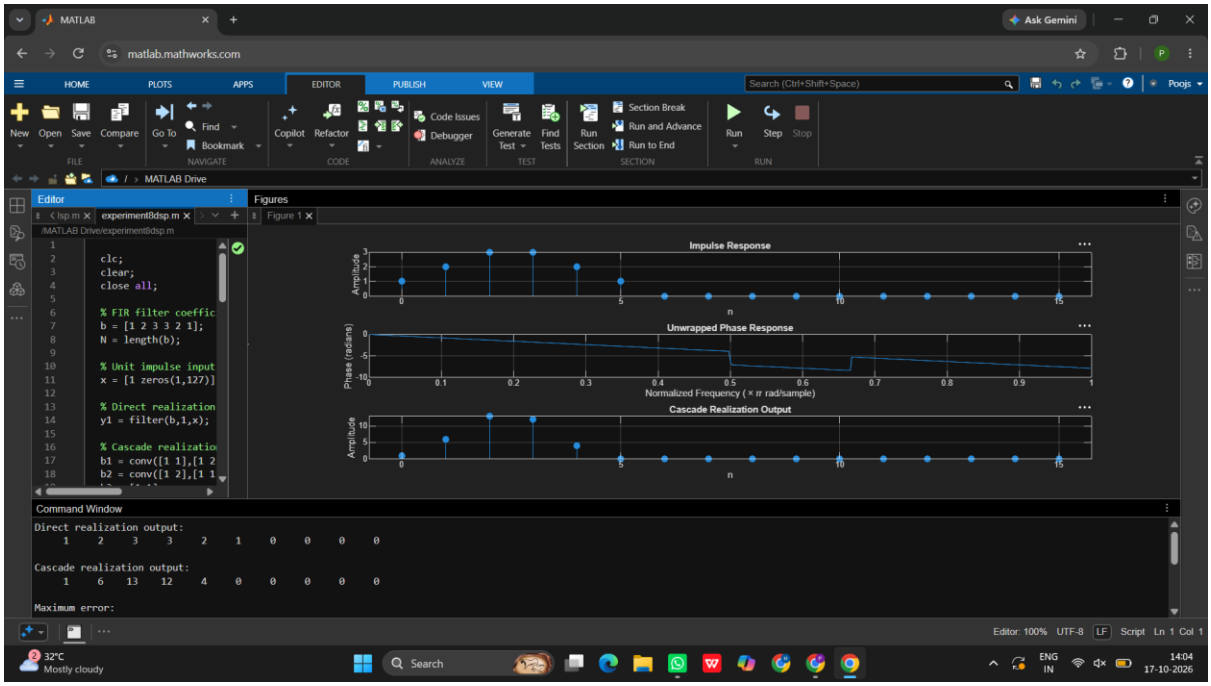
EXPERIMENT NO 8

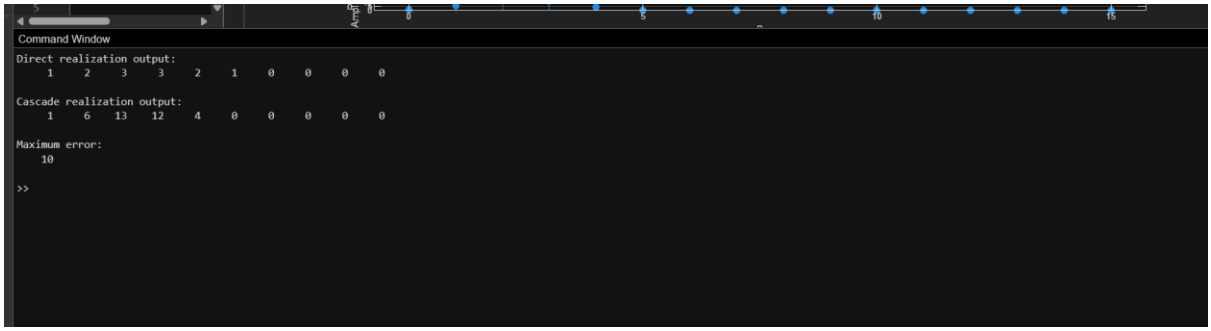
NAME: Pooja V

REG NO: 192512463

PROGRAM:

```
clc;
clear;
close all;
% FIR filter coefficients
b = [1 2 3 3 2 1];
N = length(b);
% Unit impulse input
x = [1 zeros(1,127)];
...ylabel('Phase (radians)');
grid on;
subplot(3,1,3);
stem(0:15,y2(1:16),'filled');
title('Cascade Realization Output');
xlabel('n');
ylabel('Amplitude');
grid on;
```


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Run Section
Section Break
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Run to End
Run
Step
Stop
FILE
NAVIGATE
CODE
ANALYZE
TEST
SECTION
RUN
MATLAB Drive
Editor
lsp.m
experiment8dsp.m
/MATLAB Drive/experiment8dsp.m
clc;
clear;
close all;
% FIR filter coeffic
b = [1 2 3 3 2 1];
N = length(b);
% Unit impulse input
x = [1 zeros(1,127)]
% Direct realization
y1 = filter(b,1,x);
% Cascade realizatio
b1 = conv([1 1],[1 2
b2 = conv([1 2],[1 1
Figures
Figure 1
Impulse Response
Amplitude
n
Unwrapped Phase Response
Phase (radians)
Normalized Frequency (× π rad/sample)
Cascade Realization Output
Amplitude
n
Command Window
Direct realization output:
1 2 3 3 2 1 0 0 0 0
Cascade realization output:
1 6 13 12 4 0 0 0 0 0
Maximum error:
Editor: 100% UTF-8 LF Script Ln 1 Col 1
32°C
Mostly cloudy
Search
ENG
IN
14:04
17-10-2026


Command Window
Direct realization output:
1 2 3 3 2 1 0 0 0 0
Cascade realization output:
1 6 13 12 4 0 0 0 0 0
Maximum error:
10
>>